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Equilibrium or imbalance? Rail Transit and Land Use Mix in Station Areas

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Abstract
Although it is widely reported that rail transit has the potential to encourage higher density development, it remains unclear whether rail transit leads to more mixed urban development across station areas. This article provides rare quantitative analysis of changes in land use mix around the rail transit system in Tianjin, China through an investigation into the spatial effects of a rail transit line which cuts across both highly developed and less developed areas. By using longitudinal data over a twelve-year period (2004–2016) and by comparing the entropy-based land-use mix index, the study shows that with the operation of rail transit, land use mix has increased in formerly low-mixed station catchments, but the change is not obvious for already highly diverse areas. It also shows that a more balanced development occurs in station areas with higher land use dominance, while the leading functions are intensified in station areas with lower land use dominance. By presenting a clear picture of the spatial distribution and patterns of land use mix changes over time, this article concludes that rail transit leads to more balanced development across different station areas in the context of China's rapid urbanization. The outcome provides a base for further exploring how the planning of rail transit stations may help tackle the differentiated development in cities.

Keywords Rail transit · Land use mix · Spatial variations · Tianjin

Introduction

While urban functional division was a key planning principle in much of the twentieth-century, problems associated with the over-strict separation of land uses are evident. A mixed

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land-use pattern has increasingly become a new objective of urban development. The integration of a variety of activities within proximity provides complementary functions, which may promote active travel (Tian et al. 2017), stimulate urban vitality (Jacobs 1961; Cervero and Murakami 2009), and encourage compact development (Song and Knaap 2004). As such, land use mix has been a key dimension in many new development theories such as smart city and Transit-Oriented Development (TOD) (Manaugh and Kreider 2013).

Rail transit has long been an important driving force in land development (Cervero and Landis 1997; Levinson 2007; Xie and Levinson 2008; Zhao et al. 2018). By improving accessibility and attracting more people around stations, rail transit development provides impetus for land value appreciation and land use changes (Justin and Ann 2008; Chorus, 2009). Many studies have explored the effects of rail transit on land use near stations, most focus on land value and density (Yang and Lew 2009; Yang and Pojani 2017). As one of the "3Ds" principles of TOD (density, diversity and pedestrian-friendly design) (Cervero and Kockelman 1997), mixed or diverse land use is not only a key aspect of urban form, but also a vital characteristic of urban sustainability (Zhang, 2007). It is reported that the proper mixture of urban land uses around rail stations has the potential to raise land values, increase transit ridership, and lead to higher density development (Song and Knaap 2004; Loo et al. 2010; Ibraeva et al. 2020; Liu et al. 2020). Land use mix increases transit utilization in off-peak hours and improves the urban vitality around transit nodes. It is considered as a desirable urban quality which promotes urban livability and public health (Tian et al. 2017; Mavoa et al. 2018; Rodríguez and Kang 2020). The impact of rail transit upon land use mix, however, remains unclear: whether and how rail transit would impact the development of land use mix, equilibrium or agglomeration?

This article aims to answer this question by examining land use mix change around the rail transit system in Tianjin, China by studying the spatial effects of a rail transit line which cuts across both highly developed and less developed areas. It compared the land-use mix index based on longitudinal data over a twelve-year period (2004–2016). The remainder of this article is organized as follows. We first provide a literature review on the effects of rail transit upon land use, followed by an introduction to the methodology in Sect. 3 and data analysis in Sect. 4. The article is closed with concluding discussion.

Literature review

A wealth of literature has examined the effects of rail transit on land use. Built upon the classic bid-rent theory (Alonso 1964; Muth 1969; Mills 1972), the premise for this impact is that greater access to transit services is associated with more capital-intensive land uses and higher intensification. Rail transit improves accessibility to areas located within the transit corridor and increases their relative advantages. It is expected that activities that would have occurred elsewhere will be redistributed to areas along rail corridors, and lead to increased land values and changed land use patterns in these areas (Giuliano 1986; Wang et al. 2019). Others have pointed out that such processes will only take place when conditions are conducive to development, which may involve factors such as supportive local government policies, the availability of land for development, the market attractiveness of station sites, and economic conditions (Knight and Trygg 1977; Huang 1996).

Studies have shown that transit development may increase land use intensification. Land parcels surrounding rail transit stations often have higher density than those further away, measured by floor area ratio, employment density, or population density. For example, newly built metro station areas are found to have greater population increase than areas near old stations or without stations nearby in Madrid (Calvo et al. 2013). Chorus (2009) shows that Tokyo's railway is the driving force behind the high density of city centers and sub-centers. Certainly, it should be noted that whether and to what degree such land use intensification would occur depends on local zoning policies and other urban conditions. For example, higher intensity development along Toronto's Yonge Street subway line was achieved as a result of a range of policies adopted by the Metropolitan Toronto government: changing zoning to permit high-rise development, allowing liberal floor/area ratios up to 12, awarding density bonuses, and so on (Cervero 1986). Little development may occur if station sites do not have land that can be developed (Cervero 1985) or if local governments adopt downzoning in response to neighborhood opposition (Knight and Trygg 1977). A more recent study by Xiaoguang Wang in Shenzhen, China shows that the densifying of land development brought on by rail transit is more prominent in already developed central areas than in low-density suburbs (Wang et al. 2019).

Some scholars have explored the impact of transit development on land use changes. Green and James (1993) show that growth in service employment favored Metro corridors and stations in Washington, DC, while the tendency was not consistent in retail and government. A study by Hurst and West (2014) on land use changes within the corridor of 0.5 miles of Minneapolis's Hiawatha (Blue) line shows that proximity to stations increases the likelihood of land use change in comparison to land further away.

As the interaction between rail transit and land use is affected by multiple factors, researchers find that the connection varies across buffer zones, station areas, and cities. By comparing the land use data in two buffers (0-200 m, 200–500 m) of metro stations in Shanghai, Pan and Zhang (2008) suggest that higher intensity and more capital-intensive development occur in the inner buffer. Guerra's 2014 study on the effects of the Line B Metro expansion in Mexico City shows that rail transit increased density around stations but had little to no effect on downtown commercial development. In contrast, studies of the San Francisco Bay Area Rapid Transit (BART) reveal that BART has had varied impacts on land use, with most notable land-use impacts in downtown San Francisco (Dyett et al. 1979; Cervero and Landis 1997) show that BART affected land uses only when it was supported by zoning, market demand, and community, while little impact was found in areas with deteriorating housing conditions or antidevelopment sentiment. The study by Zhao et al. (2018) shows that land use types and the level of land intensification were unrelated to metro development in Beijing; instead, the road system had a greater influence on the former. Bying using Seoul Metro Line 9, Lee (2022) explores the differences in the effects of rail transit investment on land use, across various land types, values and locations.

Only a small number of researchers have studied rail transit's effects upon land use mix. Bhattacharjee and Goetz (2016) show a modest growth of various types of land use near existing and proposed rail transit lines in Denver, with growth in commercial land use greater around rail transit than away from it. Based on analysis of parcel-level land-use data in Houston in 2005–2014, Lee and Sener (2017) also reveal a significant increase in commercial development along the light rail corridor 4 to 10 years after its opening.

In summary, while many studies have examined the correlation between rail transit and land use change by using a variety of methods, evidence on the impact of rail transit remains mixed and its effects upon *land use mix* is poorly represented in the literature. Moreover, there is little evidence that relates to cities in `developing' countries, where mixed land use is relatively more common than those in more `advanced' societies. This article contributes to existing literature by providing rare quantitative analysis of the spatial distributional effects of land use mix change in station areas over time through a longitudinal study, and by focusing on an under-studied city in China.

Method and data

Research area

China has experienced rapid urban rail development in the past two decades. By the end of 2018, 35 Chinese cities have completed the construction of 5,761 km of urban rail routes (Xinhua News, 2019). Situated southeast of Beijing, Tianjin has an administrative area of 11,966 km^2 and a population of 15.6 million as of 2019. The first metro line - "old M1 Line" was opened in the city in 1984, which was 7.4 km with eight stations. Its operation was suspended in 2001 and re-opened as the new M1 in 2006 after a four-year upgrade.

Given the time lag between rail transit development and its effects on land use (Cervero and Kang 2011; Cervero and Landis 1997), we chose the new M1 line as our case study, which was in operation for the longest time and the impact of rail transit on land use should be evident by this point. M1 travels from the Liuyuan Station in the northwest to the Shuanglin Station in the southeast of the city, with 22 stations and a total length of 42 km. There are three road rings in the central city of Tianjin: Inner Ring, Middle Ring, and Outer Ring, which divide the city into three zones. The zone within the Inner Ring is the inner core of the city, which features high-density development. The zone between Inner and Middle Ring features good public service facilities and extensive land use mix, while most parts of the zone between the Middle and Outer Ring are newly developed areas, often with a singular function. M1 crosses all three zones, with stations in six out of the ten municipal districts in and around the central city, where the economic and urban development levels differ from each other. The different locations and associated urban conditions of each station offer the opportunity to observe the diverse effects of rail transit on land use mix. It should be noted that M1 was planned at a time when TOD principles were not used in urban and transportation planning in Tianjin. Therefore, it presents an excellent opportunity to observe the effects of rail transit on land use with little policy intervention.

To evaluate the effects of the rail transit, areas away from the station catchments are selected as control sites to make a comparison of land use changes. During 2004–2016, five rail transit lines were in operation and one was under construction. With six rail transit lines in the central city of Tianjin, the 1,000 m catchments of their stations cover most of the central city. So the control sites are selected without the 1,000 m catchments of stations, and have similar functions with a nearby M1 station. As seen in Fig. 1, seven areas are selected as control sites.

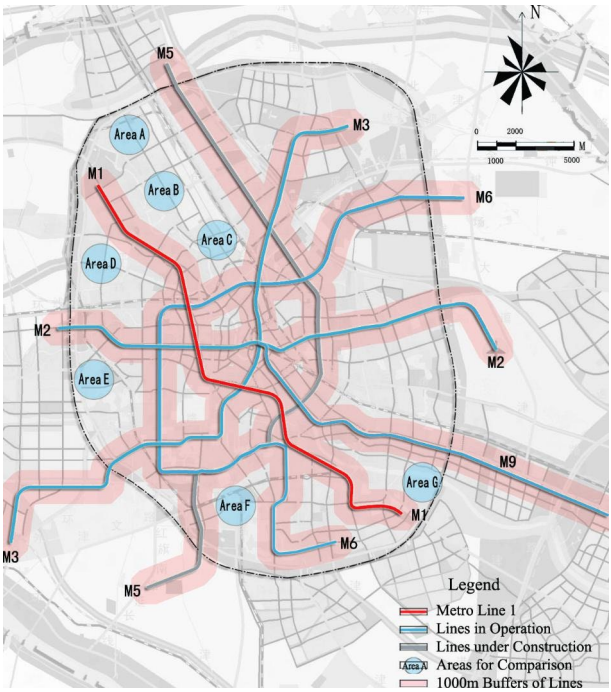


Fig. 1 Metro Corridors and Comparison Areas

Research approach

The study sought to determine whether and to what extent rail transit has impacted land use mix in station areas with the following steps. We started by defining the catchment of rail transit stations. A before-and-after approach was then applied to detect the changes in land use mix over a twelve-year period for each station. Finally, a K-means clustering analysis was introduced to classify the impact of rail transit on land use mix.

Definition of the rail transit station catchment

The impact area of rail transit station is usually delineated as within a walking distance from stations (Guerra et al. 2012; Vale 2015; Lyu et al. 2016). However, due to the lack of fixed standard for the walking distance, a variety of case-specific distances have been used. American scholars used a range from 1/4 mile (400 m) to 1/2 mile (800 m) from transit stops (Atkinson-Palombo and Kuby 2011; Schlossberg and Brown 2004), while some European researchers used a 700 m radius from transit stations instead (Reusser et al. 2008; Vale 2015; Zemp et al. 2011). Given the different demographic and geographical characteristics, some studies selected the 1,000 m threshold to define the metro catchment in Chinese cities (Pan and Zhang 2008; Yang and Liang 2013). A survey by Tianjin Urban Planning & Design Institute shows that more than 80% people in the city walk to transit stations, with nearly 80% within 10 min (Tianjin Urban Planning & Design Institute, 2012). Based on these facts, many urban design projects in Tianjin select the 1,000 m threshold to define the metro catchment. Following this local convention, this study delineated the region within a 1,000 m

radius surrounding rail transit stations as the impact area, and classified the area into two buffers: inner (0-500 m) and outer (500-1,000 m).

Calculation of land use mix indexes

There are many approaches to the measurement of land use mix that have emerged over recent decades, among which, the entropy-based land-use mix (*H*) index is the most widely used (Kockelman 1997; Ewing and Cervero 2010; Song et al. 2013).

The equation used in H was derived from Shannon (1948) in the field of communications. The index was first introduced into urban study field by Wilson (1969) to estimate traffic distribution between zones. Cervero (1989) adopted the equation to calculate the land-use mix in suburban areas of America. Based on the entropy formula, Frank and Pivo (1994) measured the land use mix by GIS, and reported a positive relationship between mixed use and travel mode choices. After being standardized by Maria Kockelman (1997) using ln(k), the index became an archetype for determining land use mix and be extended to some other fields, such as landscape ecology, real estate economy, health and preventive medicine.

The following equation was used in this study:

$$H = -\sum_{i=1}^{k} \frac{P_i \ln(P_i)}{\ln(k)} \tag{1}$$

Where:

k: the number of land-use categories.

P_i: the proportion of land area for the use of *i*.

The above index focuses on developable land. To explore whether the urban rail transit construction will bring about development and redevelopment around stations, we employ *N*, the proportion of land not permitted for development in the current Master Plan within the transit station catchment, which is identified as follows:

$$N = \frac{S_n}{S_{sum}} \tag{2}$$

Where:

N: the proportion of land not permitted for development.

S_n: the land area of land not permitted for development.

S_{sum}: the land area of the catchment.

Classification of mixed degrees

Based on *H* and *N* above, the degree of land use mix of each transit station was calculated. To examine the impact of rail transit on land use mix, we applied a before-and-after approach to observe the changes of mixed degrees over a twelve-year period. As the land value appreciation brought about by rail transit occurs after the transit planning is announced (Atkinson-Palombo 2010; Henneberry 1998; Knaap et al. 2001), and there is usually a time lag for its effects on land use development (Cervero and Kang 2011; Cervero and Landis 1997), the

research period in this article includes two years before the rail line was opened and ten years after that, to examine the long-term impacts.

To explore the heterogeneity of the impact of rail transit on land use mix, a K-means clustering analysis was introduced. The main aim of clustering analysis is to classify data into groups (clusters) with similar characteristics (Fraley and Raftery 1998). K-means clustering (MacQueen 1967) is among the most widely used clustering analysis methods. K-means clustering analysis seeks to partition the n individuals in a set of multivariate data into K clusters. As a hard partitioning algorithm, K-means clustering analysis is an iterative process. It proceeds by selecting K initial cluster centers and refining them based on the minimization of a performance index (Han et al. 2004; Wagstaff et al. 2001).

In this study, we employed the K-means clustering method to classify the transit stations into different groups based on the variations of land use indexes in order to distinguish heterogeneous effects of rail transit upon station areas with different existing urban conditions.

Research data

The study object is Metro line 1 of Tianjin, which was opened in 2006. The land use data at the parcel level between 2004 and 2016 were collected from Tianjin Urban Planning & Design Institute, including land utilization maps of the central city and the land area data of each land use type. According to the "Code for classification of urban land use and planning standards" of China (GB 50137−2011), the land use types are classified into two parts: developable land and land not permitted for development. The former is divided into eight land use types: residential, administrative and public services, commercial and business, industrial and manufacturing, logistics and warehouses, roads and transportation, municipal utilities, and green spaces (squares included). Land not permitted for development (S_n) refers to land that will not be used for construction during the specified period in the master plan. If approved by the updated master plan, land not permitted for development can be converted to developable land, while the latter can also be converted to the former if low efficient or inappropriate land use is detected. For the purpose of this study, we divided the "commercial and business" land use type into two - commercial and office - and integrated the "industrial and manufacturing" and "logistics and warehouses" into one - industrial. Meanwhile, both land use types "roads and transportation" and "municipal utilities" are integrated into one, infrastructure, which was not used to calculate the land use mix in this study. As a result, six land-use types were selected to calculate the P_i in Entropy, namely residential, commercial, office, public services, industrial, and green spaces. The data of land not permitted for development is used in N.

The location data of transit line and stations were assembled from the transit operator website. By using GIS the maps of transit line and land use of 2004 and 2016 were integrated respectively, with which the degrees of land use mix within 0-500 m and 500-1,000 m buffers of transit stations were calculated. Table 1 shows the Entropy and N of Metro Station Catchments.

Table 1 The Entropy and N of Metro Station Catchments

Stations	Inner Buffer (0-500 m)				Outer Buffer (500–1000 m)			
	N-Inner		H-Inner		N-Outer		H-Outer	
	2004	2016	2004	2016	2004	2016	2004	2016
Overall Mean Values	0.0799	0.0702	0.5457	0.5527	0.1182	0.0806	0.5546	0.5863
Liuyuan Station	0.2549	0.0025	0.4151	0.5765	0.2749	0.0816	0.4385	0.5531
Xihengdi Station	0.0200	0.0422	0.3222	0.4763	0.0830	0.0736	0.3739	0.4342
Guojiuchang Station	0.0302	0.0559	0.4187	0.3653	0.1065	0.1603	0.4381	0.4831
Benxilu Station	0.0270	0.0199	0.5353	0.4278	0.1424	0.1589	0.6077	0.6162
Qinjiandao Station	0.0780	0.0000	0.6025	0.6633	0.0848	0.0359	0.5247	0.5054
Honghuli Station	0.0345	0.1078	0.6481	0.5336	0.1745	0.0530	0.5917	0.5775
Xizhan Station	0.1339	0.0982	0.4810	0.4210	0.2293	0.1464	0.4917	0.4923
Xibeijiao Station	0.0710	0.0023	0.4042	0.5528	0.0547	0.0335	0.5018	0.5417
Xinanjiao Station	0.0200	0.0031	0.6717	0.5624	0.0906	0.0137	0.5544	0.5871
Erweilu Station	0.0311	0.0225	0.6867	0.7266	0.0731	0.0068	0.6123	0.6229
Haiguangsi Station	0.0144	0.0022	0.7058	0.6742	0.0394	0.0018	0.5846	0.6118
Anshandao Station	0.0080	0.0098	0.6278	0.6207	0.0135	0.0025	0.6103	0.6405
Yingkoudao Station	0.0145	0.0000	0.6415	0.7227	0.0251	0.0084	0.6359	0.6886
Xiaobailou Station	0.1015	0.0461	0.5984	0.7200	0.1374	0.0754	0.6329	0.7129
Xiawafang Station	0.0526	0.0180	0.6356	0.6564	0.0518	0.0693	0.5534	0.5742
Nanlou Station	0.0326	0.0158	0.5556	0.5851	0.0406	0.0747	0.6623	0.5839
Tucheng Station	0.0800	0.1138	0.6702	0.6444	0.0884	0.0606	0.6331	0.6438
Chentangzhuang Station	0.1304	0.3992	0.5559	0.3876	0.2147	0.2375	0.4841	0.5367
Fuxingmen Station	0.2387	0.3985	0.5165	0.3776	0.2026	0.1712	0.5872	0.6072
Huashanli Station	0.0700	0.0955	0.6066	0.6107	0.1000	0.1741	0.6301	0.6274
Caijingdaxue Station	0.0000	0.0000	0.4539	0.4825	0.0981	0.0208	0.5365	0.6524
Shuanglin Station	0.3135	0.0915	0.2510	0.3716	0.2744	0.1134	0.5158	0.6050

Results and discussion

Our data analysis is divided into three parts. First, by using the entropy and N in 2004 and 2016, the overall mixed-use levels were described. Second, to determine if and how rail transit might be influencing land use mix, we compared land use changes near the rail transit with those away from it to detect differences. Third, a K-means clustering analysis was introduced to explore the heterogeneity of the impact of rail transit on land use mix.

Overall mixed-use degrees

The proportion of land not permitted for development (*N*)

As the first rail transit line in Tianjin, M1 passes through the core development area of the central city, connecting the main commercial and business centers and transportation hubs to alleviate the increasing traffic pressure. Therefore, the proportion of land not permitted for development along the line was not high in 2004 (Fig. 2); only three stations located at the end of the line were found with the proportion over 20% (Table 1). Within the inner buffer, 73% of the transit stations had land not permitted for development below 10%, and the ratio was 60% in the outer buffer.

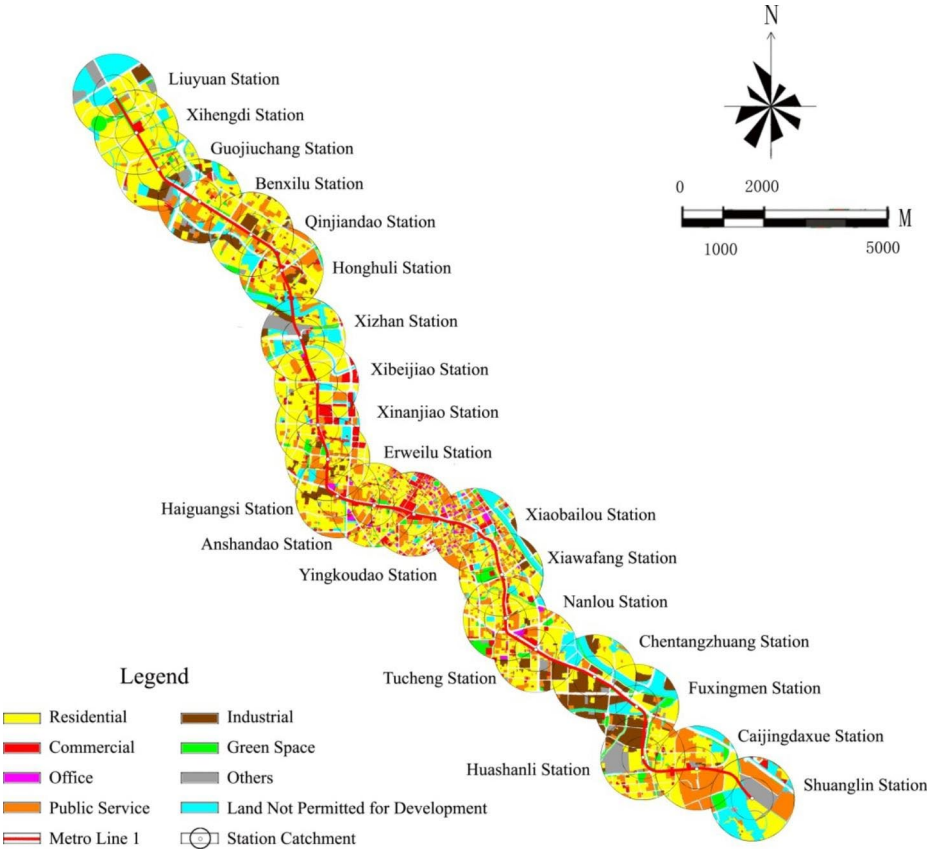


Fig. 2 The land use of M1 in 2004

With the operation of M1, the station catchment was considered by developers an obvious value appreciation due to the improvement in transit accessibility, and has thus become a hot spot for urban development and redevelopment. During the past 12 years, the land initially not permitted for development has gradually transformed to legally developable land (Fig. 3). The *N* values of all stations, except for Chentangzhang Station, Fuxingmen Station, Tucheng Station and Honghuli Station, were all below 10% in the inner buffer in 2016 (Table 1). It shows that the construction of rail transit indeed led to the development of land along the line, and the conversion of land not permitted for development in the outer buffer was higher. Within the 1,000 m radius area along M1, all the land not permitted for development was 4,767,958 m^2 in 2004, among which 24.38% was not developable and much of it was water bodies and riverside greenways. During the past 12 years, 55.78% of the land not permitted for development was made developable land with various land uses. (It should be noted that while land not permitted for development was 4,341,634 m^2 in 2016, 51.43% of which was converted from developable land in 2004 due to low efficient land use.) The change indicates the positive and substantial effects of rail transit upon the amount of legally developable land use. Planners were presumably responding to market forces by allowing more area to be developed.

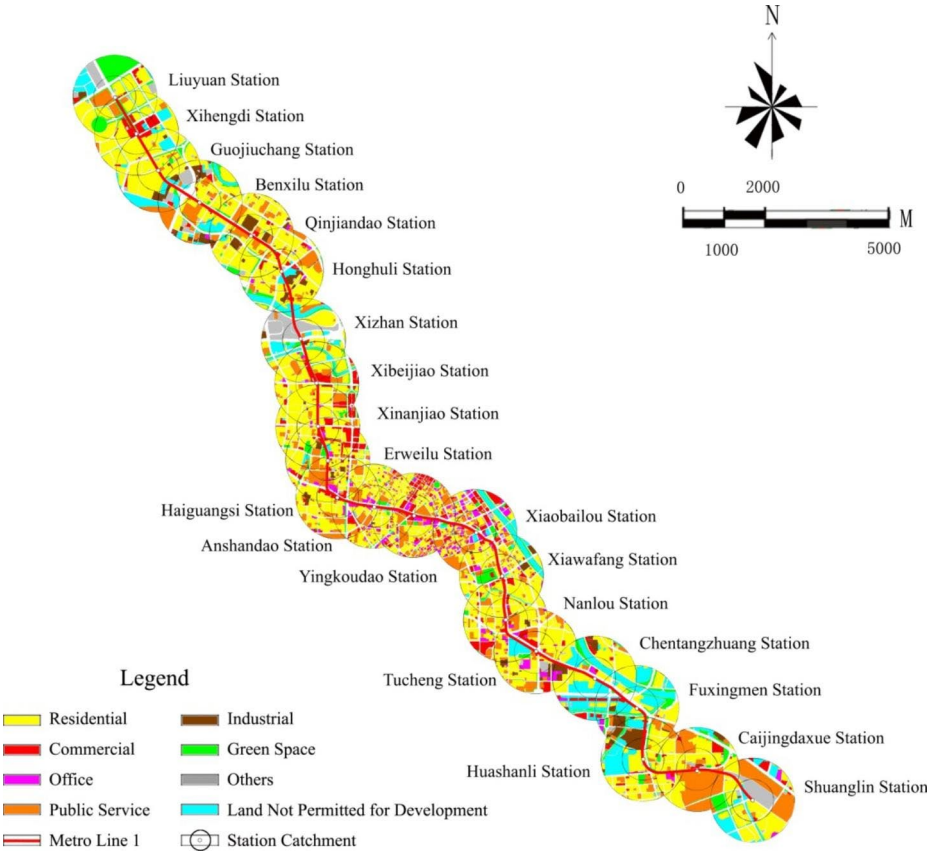


Fig. 3 The land use of M1 in 2016

The entropy index

The analysis shows a variety of land use mix values in different station areas in 2004 (Table 1). Nearly half of the stations had entropy index above 0.55 in both inner and outer buffers, with the lowest value 0.25 in the inner buffer of Shuanglin Station and the highest 0.71 in the inner buffer of Haiguangsi Station. Some studies show that the operation of a rail transit line could diversify and intensify the land use surrounding the stations (Bocarejo et al. 2013; Ratner and Goetz 2013). Yet our study reveals that it is not always the case. In the outer buffer, most of the stations show an increased trend of entropy values, while the stations with decreased values were mainly those with the entropy values above 0.55 in 2004. In contrast, in the inner buffer, the numbers of stations with increased or decreased entropy values were nearly the same. Most of the top five stations in the entropy ranking of 2004 showed a trend of decreasing in entropy values in 2016, while the majority of the bottom five stations showed an ascendant one. It demonstrates that the opening of a rail transit line could enrich land use diversity in the previously low mixed-use station catchments, but the effect was not obvious for the already highly mixed areas.

Table 2 Indexes variations of Areas and Station Catchments

	Inner Catchment		*Outer Catchment*	
Stations	*N*	*H*	*N*	*H*
Area A	-0.0992	0.1404	0.0305	-0.0964
Liuyuan Station	-0.2524	0.1614	-0.1933	0.1146
Area B	-0.0339	-0.0523	-0.0136	-0.1276
Xihengdi Station	0.0222	0.1541	-0.0094	0.0603
Area C	-0.0339	-0.0306	-0.1040	-0.0627
Xizhan Station	-0.0357	-0.0600	-0.0829	0.0006
Area D	-0.0250	0.0375	-0.1930	-0.0900
Guojiuchang Station	0.0257	-0.0534	0.0538	0.0450
Area E	-0.0694	0.0279	0.0237	-0.0196
Xibeijiao Station	-0.0687	0.1486	-0.0212	0.0399
Area F	-0.1050	0.0994	-0.0512	-0.1130
Caijingdaxue	0.0000	0.0286	-0.0773	0.1159
Area G	-0.1274	0.0282	-0.0780	-0.1747
Shuanglin Station	-0.2220	0.1206	-0.1610	0.0892

Comparative analysis

Due to the high coverage of rail transit stations in the central city of Tianjin, the selected comparative areas are mainly located in peripheral areas. In 2004, the proportion of land not permitted for development is relatively high, and the land use mix is very low. In order to isolate the impact of rail transit on land use changes, we selected the bottom seven stations in the entropy ranking of 2004 for comparison, and the station catchments with similar location and entropy value as the seven areas are in pair.

It can be seen from Table 2 that, most of the seven areas and the station catchments selected for comparison show a similar trend for reduced *N* as both the 500 m and 1,000 m catchment. But as to entropy, all the station catchments except two in the inner buffer present increased values, while all the seven areas with reduced ones in the outer buffer.

Within the 500–1000 m range, the proportion of land not permitted for development in Area A and Liuyuan Station in 2004 was 39.40% and 27.49% respectively, but 42.45% and 8.16% in 2016. The change in Area A is about 3%, while a significant decrease of nearly 20% was found at Liuyuan Station catchment. The main land use changes at Liuyuan Station were the shifts from industrial to residential, public services and green spaces, while Area A is still waiting for the redevelopment.

We see similar trends as we compared Area G and the inner buffer of Shuanglin Station: with reduced *N*, and increased entropy, but the variation value of entropy of Shuanglin Station inner buffer was over 4 times of that of Area G. The *N* values of Area G were about half of Shuanglin Station in both of the two buffers. In the outer buffer, the entropy value of Area G deceased greatly, which was contrary to the change trend at Shuanglin Station. A closer look into the land use changes indicates that while Area G also underwent redevelopment, changes mainly led to residential.

Although the variation values of *N* are both not significant for Area B, D, E, F and the corresponding station catchments, the pattern of land use change is completely different. The proportion of land not permitted for development in Area B, D, E and F in 2004 was all very high as Area A and G, but the redevelopment speed is even slower than the latter. Therefore, the values of N and entropy of the areas present sight fluctuation. As different

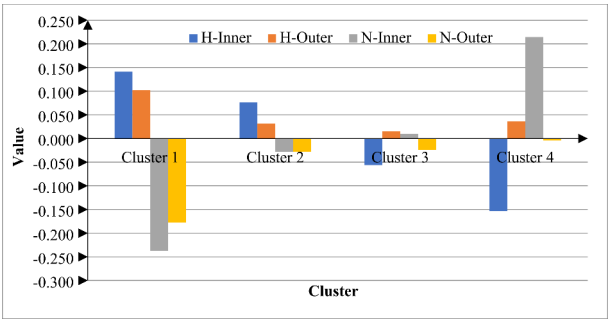


Fig. 4 The results of K-means clustering

to Area B, D, E and F, the *N* values of the corresponding station catchments, Xihengdi Station, Guojiuchang Station, Xibeijiao Station and Caijingdaxue Station were all below 10% in 2004, but the renewal is ongoing. Residential and industrial land with low land use efficiency and low quality has been transformed into commercial, public service and green space. Therefore, the land use mix of the four station catchments is increasing gradually.

As to Area C and the corresponding Xizhan Station, the same trend is reduced *N*, and reduced *entropy*. As an important transportation hub in Tianjin, the reconstruction and expansion of Xizhan Railway Station was officially launched in 2009. But the land collection in the surrounding areas started earlier, so there was more non-development land in the catchment than other stations in 2004. But with the reconstruction of the area, more functions of transportation hub is gathering there, so the *entropy* value of Xizhan Station is a bit decreased in the inner buffer.

In conclusion, the above comparisons reveal the obvious differences in land use changes near and away from the rail transit system, despite similar locations and former entropy value. The findings demonstrate that there is a correlation between rail transit and changes in land use mix.

Heterogeneity analysis

The K-means clustering analysis is applied to further identify the homogeneous subgroups of M1 station catchments, with four components of the variations of *entropy* and N as a criterion. All of the 22 stations are tested with them, and classified into four groups (Fig. 4). Table 3 shows the ANOVA analysis of K-means clustering analysis. Mean values per cluster for each factor suggest wide variation in catchment types. Some catchment types are very unique, having only two station represented (cluster 1 and cluster 4), while others represent a fairly large number of catchments (cluster 2 and 3 both consists of 9 stations). Overall, the clusters embody different combinations of relevant dimensions of rail transit's impact on land use mix.

Group 1

The first group consists of two stations, Liuyuan Station and Shuanglin Station, both being the end stations of M1 located in the Middle-Outer Ring Zone of the central city. As the less developed areas of the city, the proportions of land not permitted for development in their catchments were high: between 25% and 32% in 2004. As the rail transit line started

Table 3 ANOVA Analysis of K-Means Clustering Analysis

Variables	Mean square		F	Sig.
	Cluster	Error		
Variations of N-500	0.070	0.001	52.437	0.000
Variations of H-500	0.055	0.002	24.856	0.000
Variations of N-1000	0.015	0.003	5.540	0.007
Variations of H-1000	0.004	0.002	2.468	0.095

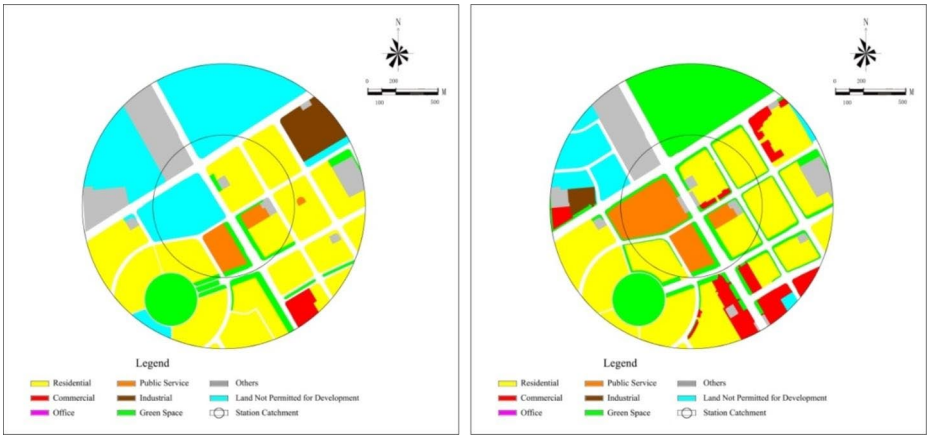


Fig. 5 The land use of Liuyuan Station in 2004 and 2016

to operate in 2006, over 60% of the land not permitted for development has been converted to developable land 2004–2016. Nearly 50% of developable land in Liuyuan Station catchment (Figs. 5) and 30% of Shuanglin Station catchment was residential in 2004, with only a small part of land for commercial and public services, which explained their low entropy ranking in 2004. The entropy value of Shuanglin Station in the inner buffer is only 0.25, which was the lowest of all the stations. With the opening of M1, much of land not permitted for development was transformed into new land use such as commercial, public services and green spaces. The entropy values of the two stations increased significantly.

Group 2

The second group consists of nine stations, namely Xihengdi Station, Qinjiandao Station, Xibeijiao Station, Erweilu Station, Yingkoudao Station, Xiaobailou Station, Xiawafang Station, Nanlou Station and Caijingdaxue Station. The proportions of their land not permitted for development were all below 10% in 2004, except for Xiaobailou Station at 10.15% and 13.74% in the two buffers. Within 12 years, the *N* values of all station catchments in this group dropped slightly, except for Xihengdi Station in the inner buffer and Xiawafang Station and Nanlou Station in the outer buffer, with a negative change ratio. The entropy index of the group showed an increased trend, with a higher appreciation in the inner buffer than the outer one. The findings show that the rail transit has led to a more balanced land use pattern surrounding stations of this group, and the effect has been more obvious in areas closer to the stations.

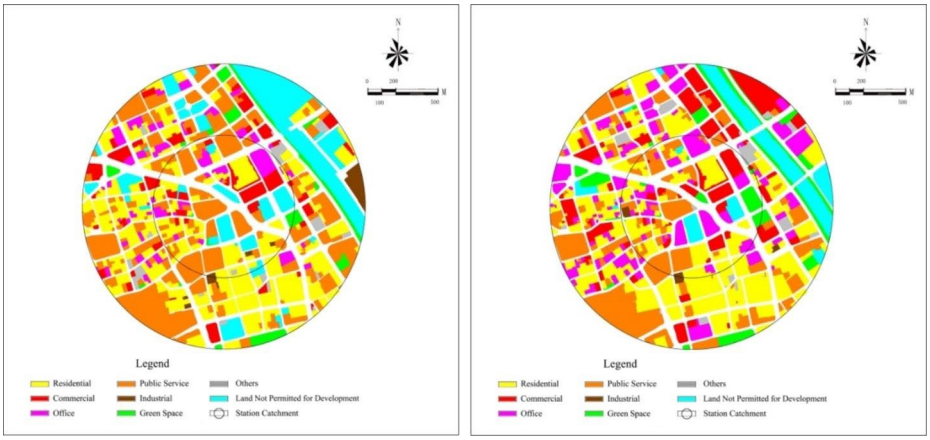


Fig. 6 The land use of Xiaobailou Station in 2004 and 2016

Erweilu Station, Yingkoudao Station and Xiaobailou Station are all in the inner city area. With high diversity of land uses, the three stations covered all the six types of development land in 2004. The entropy values were 0.69, 0.64 and 0.60 in the inner buffer, and 0.61, 0.64 and 0.63 in the outer, which were among the top six in entropy ranking. With the opening of the rail transit, the more capital-intensive land use types, such as commercial and office, were found to be concentrated in the inner buffer (Fig. 6).

Group 3

Nearly half of the stations are of this group, which can be further divided into two subgroups based on their location and development level. Subgroup 1 consists of three stations, namely Xinanjiao Station, Haiguangsi Station and Anshandao Station, all of which are located in the Inner Ring Zone. Subgroup 2 has six stations, namely, Guojiuchang Station, Benxilu Station, Honghuli Station, Xizhan Station, Tucheng Station, and Huashanli Station, which are located in either the Inner-Middle Ring Zone or Middle-Outer Ring Zone. The proportions of land not permitted for development at all station catchments were all below 20% in 2004, except Xizhan Station. The *N* values of Subgroup 1 all decreased within the two buffers in 2016, while most stations of Subgroup 2 showed an upward trend in the inner buffer and downward trend in the outer one. This could be explained by the fact that while rail transit encouraged redevelopment, the scale of redevelopment in core areas was much larger than that of peripheral areas.

The land use changes showed different trends in the two buffers. Most of the stations had an increase around 2% in the outer buffer, while a downward trend up to 5% in the inner buffer. This indicates that for the stations with existing high degree of land use mix, the effect of rail transit on land use mix was not obvious.

Group 4

The last group consists of two stations, namely Chentangzhuang Station and Fuxingmen Station, both of which are located in the Middle-Outer Ring Zone. The proportions of their

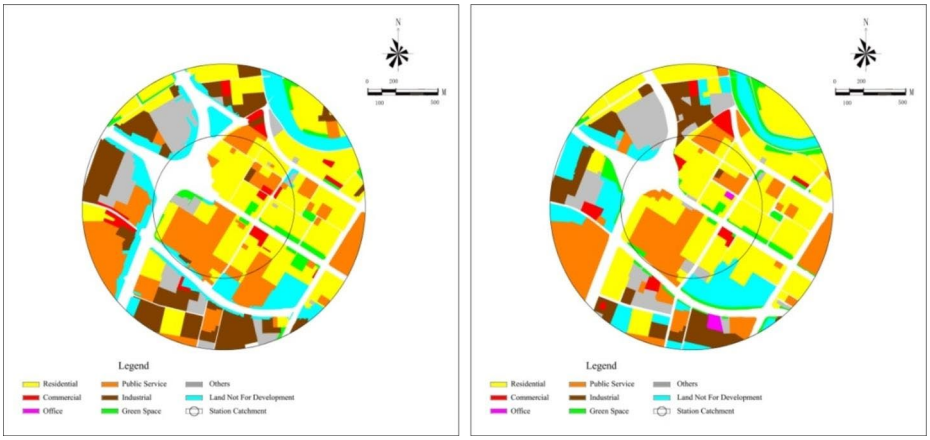


Fig. 7 The land use of Benxilu Station in 2004 and 2016

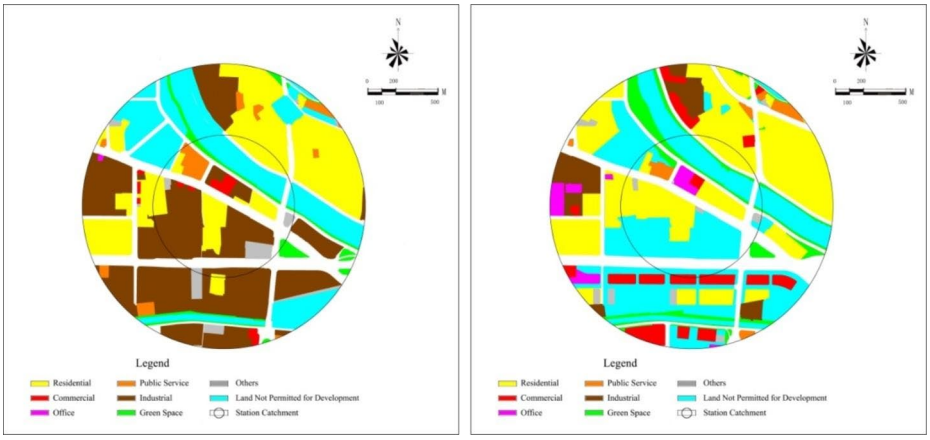


Fig. 8 The land use of Chentangzhuang Station in 2004 and 2016

land not permitted for development were between 10% and 25% in 2004, and have gone upward greatly in the past 12 years. The *N* values of Chentangzhuang Station (Fig. 8) have increased 206% in the inner buffer and around 11% in the outer zone. Chentangzhuang Station and Fuxingmen Station are located adjacent to Haihe, the mother river of Tianjin, and most of land not permitted for development in their catchments are water bodies, greenways along the river and land to be redeveloped. With the opening of rail transit, the two stations went through the processes of development and redevelopment. Yet while some industrial functions moved to other areas, the redevelopment process has not yet started, resulting in a substantial increase in the proportion of land not permitted for development.

Conclusion and policy implications

This article provides rare quantitative analysis of changes in land use mix around the rail transit system in Tianjin by using longitudinal data over a twelve-year period (2004–2016) and by comparing the land-use mix index. Specifically, it studied the spatial effects of a rail transit line which cuts across both highly developed and less developed areas. Further, the study divided station catchments into two buffers: inner buffer (0-500 m) and outer buffer (500-1,000 m). We have also compared land use mix changes near the rail transit with those away from it. Differences were detected between the rail transit with those away from the latter, which demonstrates a correlation between rail transit and land use mix. Similar to findings in Hurst and West (2014) and Cervero and Kang (2011), the study shows that the proximity to stations increases the likelihood of land-use change. As such, the study not only allows a systematic comparative analysis of the variations in land use mix at different stages of development, but also offers a better understanding of its spatial distribution in the inner and outer buffers. The key findings and their policy implications can be summarized as follows:

First, there has been an increase in land use mix in formerly low-mixed station catchments, but the change is not obvious for the already highly diverse areas. Unlike previous studies (Bhattacharjee and Goetz 2016; Lee and Sener 2017), which show that the growth of mixed land use is not noticeable around the rail transit lines, this study reveals instead that those with an upward trend of land use mix are the ones with low degree of land use mix before the introduction of rail transit. In contrast, in areas already with a high degree of land use mix, half of the stations show varying degrees of decline in land use mix. This finding suggests that low-mixed areas should be the focal areas for the application of TOD-based planning principles for mixed development given their greater potential to enhance land-use diversity.

Second, there is a tension between agglomeration economies (low entropy) and land use diversity (high entropy): stations with relatively low entropy became more heterogeneous, stations with relatively more entropy became less heterogeneous. This finding suggests that policy makers need to differentiate their application of TOD-based planning principles between mixed development in areas with higher and lower land use concentrations. To achieve higher degree of economic efficiency after the introduction of rail transit, it would help to further diversify the land use types in the former, while encouraging agglomeration in the latter.

Third, land use mix changes differ between the inner and outer buffer, with more capital-intensive land uses, such as commercial and office, in the 500 m-radius range, and more residential in the 500-1,000 m-radius range. This finding is consistent with the classic bid-rent theory and Pan and Zhang's (2008) previous study which suggests that establishments aiming to maximize their profitability are more willing to pay more for land with better access to transit services. As the residential function is more likely to be located in the 500-1,000 m-radius range instead of the inner buffer, it is important to ensure better connectivity between stations and residential districts by creating pedestrian-friendly road networks and easy transfer between rail service and buses to residential areas.

In line with recent research on the interaction between land use mix and rail transit, this study adds new empirical evidence about changes in land use mix in station areas in a developing country setting. More significantly, for the first time it presents a clear understanding

on how changes differ between highly developed and less developed areas, between areas with more and less entropy, and between the inner and outer buffers. By presenting a clear picture of the spatial distribution and patterns of land use mix changes over time, this article concludes that rail transit leads to more balanced development across different station areas in the context of China's rapid urbanization. The outcome provides a base to further explore how the planning of rail transit stations may help tackle the differentiated development in many cities.

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Authors' contribution Yuning Wang: conception and design of study, acquisition of data, analysis and interpretation of data, drafting the manuscript, revising the manuscript. Duanfang Lu: conception and design of study, analysis and interpretation of data, revising the manuscript. David Levinson: analysis and interpretation of data, revising the manuscript.

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